BEING COHEN-MACAULAY IS LOCAL IN THE SYNTOMIC TOPOLOGY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A scheme is Cohen-Macaulay if and only if it is locally Noetherian and satisfies (S_k) for all k. Properties, Lemma properties-lemma-scheme-CM-iff-all-Sk, invoked at proof lines 5150-5152. Reduces Cohen-Macaulayness to the (S_k) conditions plus local Noetherianness.
- The (S_k) condition is local in the syntomic topology. Same chapter, Lemma lemma-Sk-local-syntomic, invoked at proof line 5149. Gives syntomic locality of the (S_k) conditions.

Source reference. Original lemma 036A: sources/stacks-descent/pinned-original-source.tex, source lines 5110–5110. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0342: sources/stacks-chow/tags.txt, source lines 4077–4077. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(S) =$ "S is Cohen-Macaulay" is local in the syntomic topology.*

Proof. This is clear from Lemma 036A above since a scheme is Cohen-Macaulay if and only if it is locally Noetherian and (S_k) for all $k \geq 0$, see Properties, Lemma 0342. □

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